

Lipoma Extraction Surgery Simulation in a Multi-user Environment

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ABSTRACT

The present work proposes the use of a multi-user virtual environment for simulation of the lipoma extraction surgery. Starting from the characterization of the procedure, all the steps were mapped to a virtual environment. The prototype implemented include two main actors, the surgeon and the medical instrumentalist, and observers that can join the simulation at any time. Voice chat was included in order to allow lifelike operating room communication, a key element for a successful simulation. Initial evaluation indicated that the proposed solution is promising in a work-from-home environment as a complementary tool for medical education.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Applied computing—Life and medical sciences; Software and its engineering—Virtual worlds training simulations

1 INTRODUCTION

Lipomas are tumors of adipose tissue located between muscle and skin [6] and their treatment usually consists of surgical removal. Physical simulators, such as biological [3] and non-biological models [4], allow the use of real instruments to practice the full lipoma removal procedure taking into account epidermis, dermis, and subdermal layer. These simulators can be deployed in a real operating room environment to increase realism but some limitations appear due to the required single-use consumables. Virtual Reality (VR) medical simulators are a solution to overcome this problem allowing practicing with a virtual patient in a controlled environment providing an engaging learning process [8]. Another advantage of VR simulators is the possibility to use a networked virtual environment, in which several users can interact at the same time expanding its use not only to surgeons but anesthesiologists, nurses, and medical instrumentalists. This lead to an improvement in communication skills between all participants.

In this paper, a multiuser environment for simulation of a lipoma extraction is presented as a potential complementary tool for medical education. The prototype development is presented in section 2 and the evaluation in section 3.

2 PROTOTYPE DEVELOPMENT

In order to develop the prototype for lipoma removal simulation, the real procedure was characterized [5]. First, a selection of required surgical instruments is done, later the location of the tumor is identified and marked, and then the surgery is performed. All the tasks performed by the involved users are identified:

- Surgeon: Performs the incision, dissection, extraction, and suturing.
- Anesthesiologist: Applies the anesthesia on the area surrounding the lipoma.
- Instrumentalist: Prepares the required instrumental and hands them to the surgeon.

For this initial prototype, a surgeon and an instrumentalist were selected, since they are involved with the main tasks of the procedure, which in some cases can be performed with local anesthesia administered by the surgeon himself. The stages of development for the prototype are presented in figure 1.

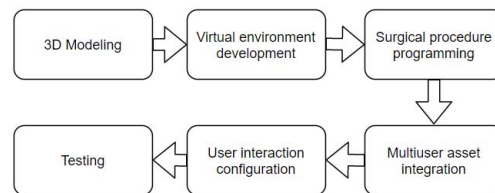


Figure 1: Procedure steps of the lipoma extraction simulation

2.1 Virtual Environment

The virtual environment was developed using a set of models acquired and designed in Blender and Adobe Fuse, including the operating room furniture, surgical instruments and equipment, and medical staff avatars (virtual patient, surgeon, and instrumentalist) as shown in figure 2. For the integration and programming Unity 3D was selected due to its compatibility with several VR equipment as well as its multi-user capabilities and large documentation.



Figure 2: Operating room virtual environment

Based on the identification of the actions performed by the actors, like instrument selection and manipulation, and patient interaction, the different steps of the procedure (surgical field preparation, anesthesia, incision, dissection, extraction, and suturing) were defined in the virtual environment as presented in figure 3.

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Each step function and its related interaction was mapped to the keyboard and mouse for the initial prototype. With this type of interaction, each user can use the prototype without the need for expensive virtual reality equipment and work from home due to the restrictions of the COVID-19 pandemic.

The avatar control permits the interaction with the environment, including movement through the OR and hand movement to manipulate required objects. Besides, animations of the instruments were implemented to increase the realism and simplify the actions carried out by the users. Audio cues were added to the basic interactions to simulate the collision between objects.

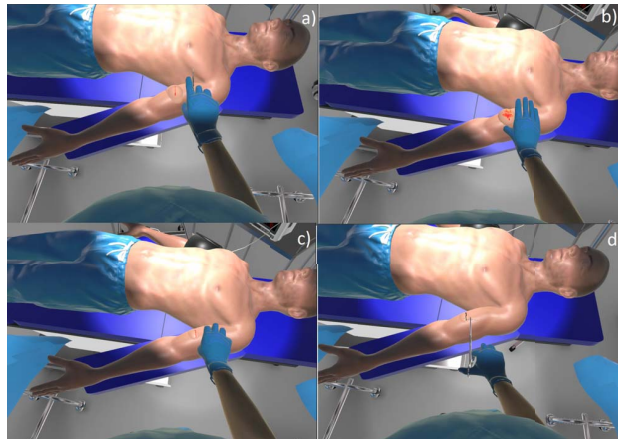


Figure 3: Lipoma removal procedure: a) Incision, b) Dissection, c) Extraction, d) Suturing

2.2 Multi-user Programming

To permit the interaction between the involved actors using the Internet, a client-server architecture was selected. Both surgeon and instrumentalist can be selected including spectators as observers of the simulation. This implementation was performed by the integration of Photon Unity Networking (PUN) [1] version 2 asset to the prototype. PUN has several free servers around the world available for the interconnection of users. The online configuration consisted of the creation of an initial menu where each user can select a role, and the transmission of movements and interactions within the environment through the server, instrumental transfer between both avatars, and voice chat communication.

The transfer of the instruments and voice chat in real-time are the most important features to simulate a lifelike operating room communication and interaction. Poor communication can cause a failure of the procedure [2].

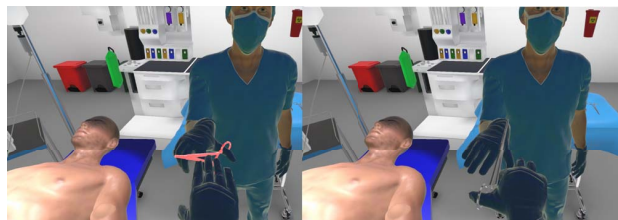


Figure 4: Instrumental transfer

3 PROTOTYPE EVALUATION

The prototype was evaluated using Nielsen 10 usability heuristics for user interface Design [7] by the development team. The main finding is the difficulty of user control and freedom related to the

movement around the virtual environment and the handling of surgical instruments. Even though the keyboard and mouse are not immersive devices like other VR haptic controllers, and both require synchronization to move in the desired way, the manipulation of the avatar's hand is possible with a little effort and practice of the user. The match between the system and the real world is supported by the voice chat and audio cues allowing each user to interact appropriately in their role for the surgery.

On the other hand, the networking performance of the prototype was evaluated in two configurations, between users in the same city and users in different countries. Obtaining a smooth performance without delays, with a correct synchronization of animations and stable voice-chat communication. This is because the PUN library has a series of functions polling the different servers for the best performance by checking the latency of the computer where the software is running. This allows a correct execution of all the tasks.

4 CONCLUSIONS AND FUTURE WORK

A prototype of a multi-user VR environment for lipoma extraction surgery simulation was presented where a surgeon and an instrumentalist can interact simultaneously. The networking tests performed between users located in different geographical locations showed that this tool has the potential to complement the training in medical education by allowing work from home or gather different people without being in the same place at the same time. Heuristic evaluation raises different challenges to improve the interaction between the user and the virtual environment.

Future work will include an in-depth usability and knowledge retention evaluation with residents of surgery to establish the real impact of the simulator. The integration of consumer-level VR equipment will be implemented to permit a higher realism interaction with the environment using hand tracking and/or haptic controllers.

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REFERENCES

- [1] Exit Games. Photon Unity Networking (PUN), 2020.
- [2] L. L. Frasier, S. R. P. Quamme, Y. Ma, D. Wiegmann, G. Levenson, E. H. DuGoff, and C. C. Greenberg. Familiarity and communication in the operating room. *Journal of Surgical Research*, 235:395–403, 2019.
- [3] V. H. Gámez-Huerta, C. D. Martínez-Hernández, and C. A. Rodríguez-Paz. Modification and evaluation of biological model of surgical teaching in extraction of lipomas in undergraduate. *Cirujano General*, 40(2):70–77, 2018.
- [4] Limbs & Things LTD. Lipoma Pad, 2020.
- [5] J. M. Martos, M. Rosa, and S. Sanz. Lipoma. *AMF*, 14(10):358–361, 2018.
- [6] Mayo Clinic. Lipoma, 2020.
- [7] J. Nielsen. Enhancing the explanatory power of usability heuristics. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*, pp. 152–158, 1994.
- [8] B. Perez-Gutierrez, A. Uribe-Quevedo, L. Vega-Medina, J. S. Salgado, N. Jaimes, and O. Perez. Immersive and non-immersive vr percutaneous coronary intervention simulation for acute myocardial infarction. In *2020 IEEE 8th International Conference on Serious Games and Applications for Health (SeGAH)*, pp. 1–4, 2020. doi: 10.1109/SeGAH49190.2020.9201902